Example 12 : Consider an electrical device that is hooked up to n batteries.

Suppose that the n batteries are independent and their lifetimes ($Y_1, Y_2, ..., Y_n$) are described by Exponential random variables with mean θ.

Consider the following two scenarios:

1. The device will function only if all batteries are functioning.
2. The device will function as long as at least one battery is functioning.

In both cases:

- Define the life of the system as a function of $Y_1, ..., Y_n$.
- Find the probability distribution for the lifetime of the device.
- How long can you expect the device to last?

6.7 : Order Statistics

Theorem 6.5a: Let $Y_1, \ldots, Y_n$ be independent and identically distributed continuous random variables with cumulative distribution function $F(y)$ and density function $f(y)$. We denote the ordered (ranked) variables by $Y_{(1)}, \ldots, Y_{(n)}$, so that $Y_{(1)} \le Y_{(2)} \le \cdots \le Y_{(n)}$.

e.g. $Y_{(1)} = minimum(Y_1, \ldots, Y_n)$ and $Y_{(n)} = maximum(Y_1, \ldots, Y_n)$.

The density of the $k-th$ ordered (ranked) variable ($Y_{(k)}$) is given by

$$f_{(k)}(y) = \frac{n!}{(k-1)!(n-k)!}\left[F(y)\right]^{k-1}\left[1-F(y)\right]^{n-k} f(y).$$

Theorem 6.5a (Proof for $Y_{(1)}$ and $Y_{(n)}$):

Example 12 : Consider an electrical device that is hooked up to n batteries.

Suppose that the n batteries are independent and their lifetimes ($Y_1, Y_2, \ldots, Y_n$) are described by Exponential random variables with mean θ.

Consider the following two scenarios:

1. The device will function only if all batteries are functioning.
2. The device will function as long as at least one battery is functioning.

In both cases:

- Define the life of the system as a function of $Y_1, \ldots, Y_n$.
- Find the probability distribution for the lifetime of the device.
- How long can you expect the device to last?

Example 13 : On a dangerous stretch of road of length 1, accidents occur independently at random ($Uniformly$) over the length of the road. At one end of the road there is an ambulance dispatch centre. Suppose it is located at position "0", and the end of the road is position "1". Let Y denote the one-way travel distance to an accident from the dispatch centre. Thus, $Y \sim Uniform(0,1)$. Suppose on a particular day there were n accidents. What is the probability distribution for the k-th longest one-way trip of the day?

Example 7 : Two individuals, Bob and Mary, agree to meet for dinner at a restaurant between 6pm and 7pm. Let X denote the proportion of time after 6pm that Bob arrives, and Y denote the proportion of time after 6pm that Mary arrives. The joint density function for X and Y is

$$f(x,y)=\begin{cases}4x(1-y) \ , & 0<x<1,\ 0<y<1 \\ 0 \ , & else\end{cases}$$

e) Find the probability distribution of $U=\min(X,Y)$.

f) Find the probability distribution of $V=\max(X,Y)$.

Example 20 : The proportion of decay for two substances are jointly distributed random variables, with joint density function given by

$$f(x,y)=\begin{cases}3y & , 0<1-x<y<1\\ 0 & , \text{otherwise}\end{cases}$$

Find the expected value for the substance with the highest decay.

<u>Theorem 6.5b</u>: Let $Y_1, \ldots, Y_n$ be independent and identically distributed continuous random variables with cumulative distribution function $F(y)$ and density function $f(y)$.

The joint density of the $j-th$ and $k-th$ ($j < k$) ordered (ranked) variables is given by

$$f_{(j),(k)}(y_j, y_k) = \frac{n!}{(j-1)!(k-1-j)!(n-k)!}\left[F(y_j)\right]^{j-1}$$
$$\times\left[F(y_k) - F(y_j)\right]^{k-1-j}\left[1 - F(y_k)\right]^{n-k} f(y_j) f(y_k).$$

Example 14 : Consider two independent $Uniform(0,1)$ random variables. What is $P(2Y_{(1)} < Y_{(2)})$?